

# Genome-wide fusion-hotspot analysis of msk\_impact\_50k\_2026

## Abstract

This manuscript synthesizes a genome-wide fusion-hotspot cohort scan of msk\_impact\_50k\_2026: 3919 genes carried at least one structural-variant record in the cohort, of which 544 passed the  $\geq 5$ -distinct-patient recurrence gate. All 544 gated genes were attempted: 4 using hand-curated gene configs and 520 auto-configured, with 20 gated-in gene(s) left unresolvable; 523 of the 544 attempted genes were successfully analyzed with the full registered algorithm suite. 4 genes reached genome-wide Benjamini-Hochberg FDR significance ( $q < 0.05$ ): NTRK3 (58 events, 94.8% in-frame, 94.8% domain-retained, Fisher  $p=0.0695489$ ,  $q=8.1653e-09$ ); ROS1 (122 events, 59.0% in-frame, 87.7% domain-retained, Fisher  $p=0.214047$ ,  $q=0.000221281$ ); ETV6 (90 events, 71.1% in-frame, 75.6% domain-retained, Fisher  $p=5.12326e-06$ ,  $q=0.00233279$ ); FLI1 (118 events, 61.9% in-frame, 5.9% domain-retained, Fisher  $p=0.988837$ ,  $q=0.00820622$ ). 15 additional genes form a highly ranked non-FDR-significant tier flagged for targeted follow-up (see Honorable mentions, below).

## Methods

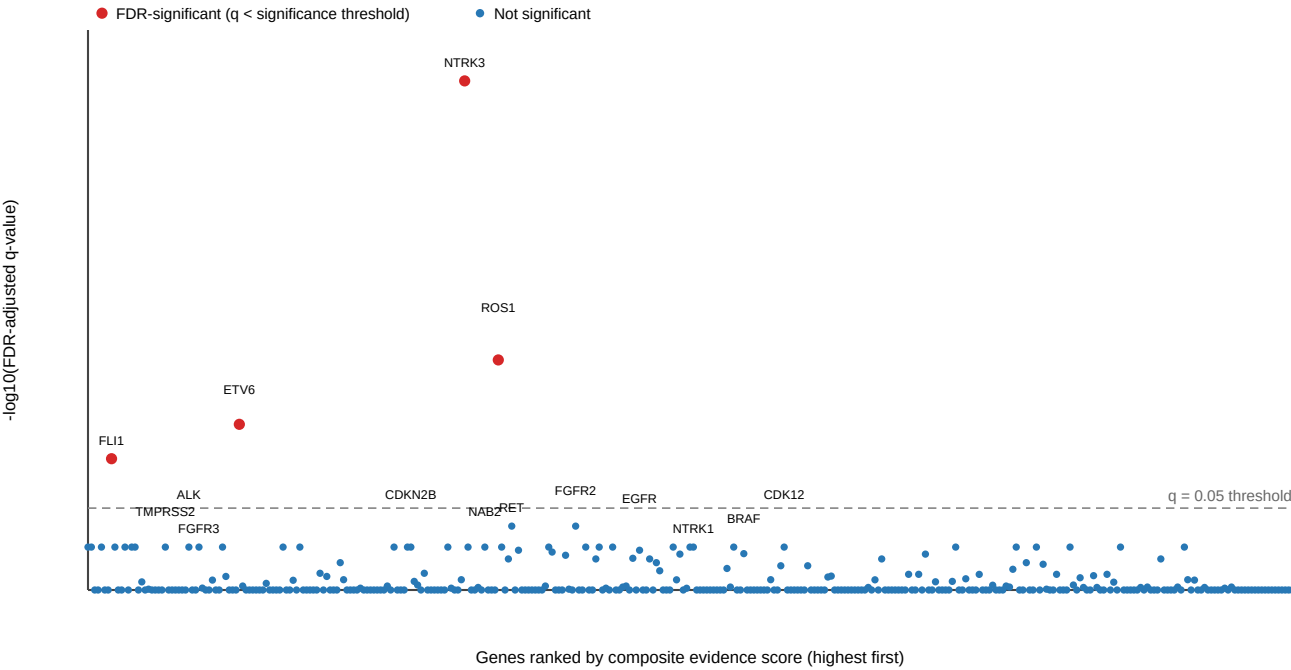
Structural-variant records were retrieved from the msk\_impact\_50k\_2026 cBioPortal study and gated to genes with at least 5 distinct patient(s) carrying a structural-variant record (544 of 3919 genes passed the gate). All 544 gated genes were attempted: 4 using hand-curated gene configs and 520 auto-configured from Genome Nexus canonical-transcript/Pfam-domain data, with 20 gated-in gene(s) left unresolvable; 523 of the 544 attempted genes were successfully analyzed. Each successfully analyzed gene was run through the algorithm suite recorded for this scan (composite\_score, confidence\_stats, cutpoint\_detection, domain\_disruption, domain\_retention, exon\_retention, frequency, joint\_partner). Domain-retention and domain-disruption significance were assessed per gene with Fisher's exact test and a breakpoint-position permutation test; the resulting p-values across the 359 genes that produced at least one computable p-value were jointly corrected with Benjamini-Hochberg false-discovery-rate correction at  $q < 0.05$ .

## Results

### Genome-wide summary

*Genome-wide summary of 359 scanned genes with an FDR-adjusted q-value, ranked left-to-right by composite evidence score; the dashed line marks the  $q=0.05$  significance threshold, with 4 genes above it.*

Genome-wide fusion-hotspot summary: FDR significance vs. composite evidence rank



## FDR-significant and honorable-mention genes

| Gene   | Tier              | N events | In-frame% | Domain-ret.% | Fisher p  | Best q    |
|--------|-------------------|----------|-----------|--------------|-----------|-----------|
| NTRK3  | FDR-significant   | 58       | 94.83     | 94.83        | 0.06955   | 8.165e-09 |
| ROS1   | FDR-significant   | 122      | 59.02     | 87.7         | 0.214     | 0.0002213 |
| ETV6   | FDR-significant   | 90       | 71.11     | 75.56        | 5.123e-06 | 0.002333  |
| FLI1   | FDR-significant   | 118      | 61.86     | 5.932        | 0.9888    | 0.008206  |
| RET    | Honorable mention | 194      | 75.26     | 92.27        | 0.0004197 | 0.09669   |
| FGFR2  | Honorable mention | 136      | 79.41     | 84.56        | 0.0004247 | 0.09669   |
| ALK    | Honorable mention | 272      | 83.09     | 95.96        | 0.001917  | 0.2081    |
| EGFR   | Honorable mention | 55       | 50.91     | 74.55        | 0.01145   | 0.2335    |
| BRAF   | Honorable mention | 179      | 84.36     | 91.06        | 0.01337   | 0.2646    |
| FGFR3  | Honorable mention | 152      | 48.68     | 93.42        | 0.01948   | 0.2081    |
| CDKN2B | Honorable mention | 12       | 16.67     | 16.67        | 0.02222   | 0.2081    |
| IKBKE  | Honorable mention | 12       | 25        | 33.33        | 0.02424   | 0.3211    |
| TPRSS2 | Honorable mention | 867      | 29.99     | 6.113        | 0.0292    | 0.2081    |
| NTRK1  | Honorable mention | 78       | 41.03     | 82.05        | 0.04826   | 0.2081    |
| INPPL1 | Honorable mention | 14       | 28.57     | 50           | 0.04895   | 0.4687    |
| CDK12  | Honorable mention | 43       | 16.28     | 37.21        | 0.0677    | 0.2081    |
| NAB2   | Honorable mention | 72       | 34.72     | 68.06        | 0.1126    | 0.2081    |
| KDM5A  | Honorable mention | 10       | 30        | 50           | 0.119     | 0.898     |
| FH     | Honorable mention | 16       | 6.25      | 12.5         | 0.1333    | 0.6863    |

## Gene highlights

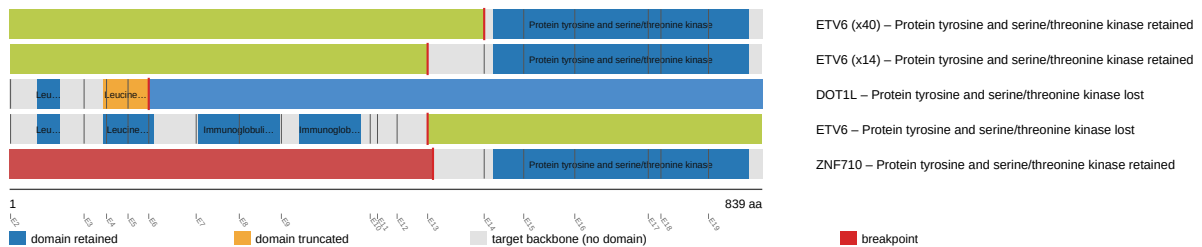
### NTRK3 (FDR-significant)

NTRK3 was analyzed across 58 fusion events, 94.8% in-frame and 94.8% domain-retained. Domain-retention Fisher's exact test  $p=0.0695489$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=8.1653e-09$  (reaches genome-wide FDR significance).

Reused from the NTRK3 individual gene report ([fusion\\_schematic.svg](#)).

NTRK3 fusion-transcript schematic

partner block → breakpoint → retained NTRK3 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/ntrk3/report.md](#)

### ROS1 (FDR-significant)

ROS1 was analyzed across 122 fusion events, 59.0% in-frame and 87.7% domain-retained. Domain-retention Fisher's exact test  $p=0.214047$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.000221281$  (reaches genome-wide FDR significance).

Reused from the ROS1 individual gene report ([fusion\\_schematic.svg](#)).

## ROS1 fusion-transcript schematic

partner block → breakpoint → retained ROS1 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/ros1/report.md](gene_reports/ros1/report.md)

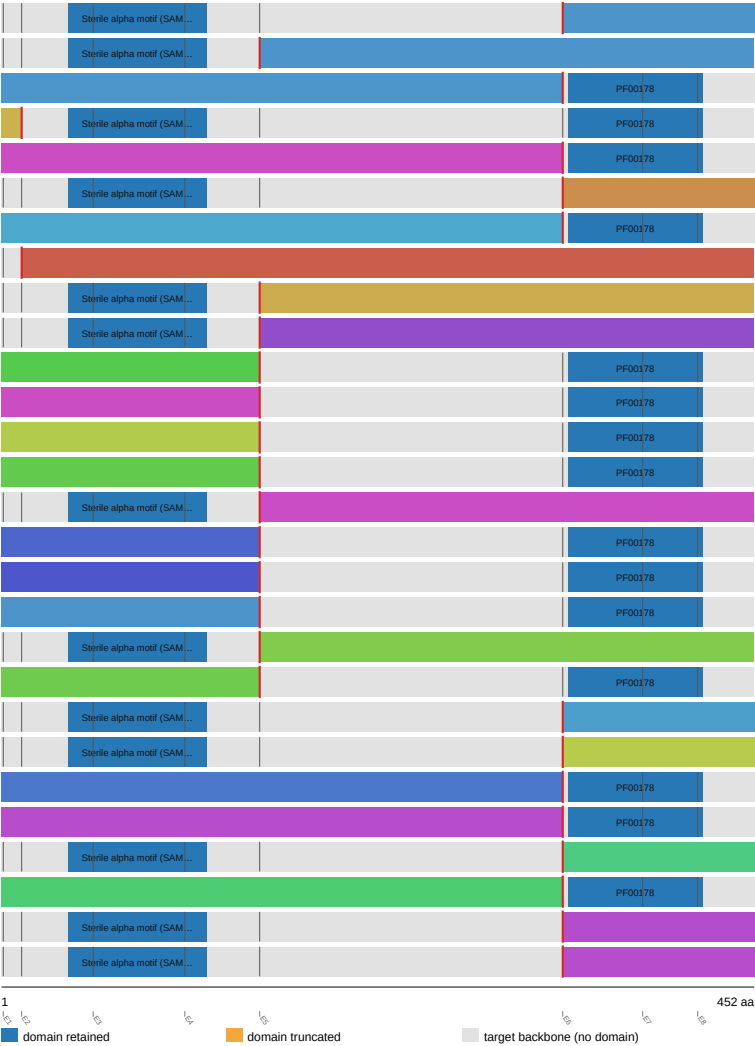
## ETV6 (FDR-significant)

ETV6 was analyzed across 90 fusion events, 71.1% in-frame and 75.6% domain-retained. Domain-retention Fisher's exact test  $p=5.12326e-06$  (raw statistically significant at  $\alpha=0.05$ ). The Sterile alpha motif (SAM)/Pointed domain appears to be required for retention. Genome-wide BH-adjusted  $q=0.00233279$  (reaches genome-wide FDR significance).

Reused from the ETV6 individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

ETV6 fusion-transcript schematic

partner block → breakpoint → retained ETV6 portion (domain-colored, exon ticks)



Showing the top 28 of 30 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

- NTRK3 (x43) – Sterile alpha motif (SAM)/Pointed domain retained
- NTRK3 (x11) – Sterile alpha motif (SAM)/Pointed domain retained
- BCL2L14 (x3) – Sterile alpha motif (SAM)/Pointed domain lost
- HCFC1 (x2) – Sterile alpha motif (SAM)/Pointed domain retained
- BORCS5 (x2) – Sterile alpha motif (SAM)/Pointed domain lost
- SGO2 (x2) – Sterile alpha motif (SAM)/Pointed domain retained
- SLIT2 (x2) – Sterile alpha motif (SAM)/Pointed domain lost
- CDKN1B – Sterile alpha motif (SAM)/Pointed domain lost
- ABCC9 – Sterile alpha motif (SAM)/Pointed domain retained
- ANKS1B – Sterile alpha motif (SAM)/Pointed domain retained
- ARHGAP26 – Sterile alpha motif (SAM)/Pointed domain lost
- BORCS5 – Sterile alpha motif (SAM)/Pointed domain lost
- CUX2 – Sterile alpha motif (SAM)/Pointed domain lost
- GALNT18 – Sterile alpha motif (SAM)/Pointed domain lost
- GNB1 – Sterile alpha motif (SAM)/Pointed domain retained
- GRIN2B – Sterile alpha motif (SAM)/Pointed domain lost
- NOL4 – Sterile alpha motif (SAM)/Pointed domain lost
- NTRK3 – Sterile alpha motif (SAM)/Pointed domain lost
- PLEKHG7 – Sterile alpha motif (SAM)/Pointed domain retained
- PTPRN2 – Sterile alpha motif (SAM)/Pointed domain lost
- AEBP2 – Sterile alpha motif (SAM)/Pointed domain retained
- APOLD1 – Sterile alpha motif (SAM)/Pointed domain retained
- FAM234B – Sterile alpha motif (SAM)/Pointed domain lost
- FGFR10P2 – Sterile alpha motif (SAM)/Pointed domain lost
- IKZF3 – Sterile alpha motif (SAM)/Pointed domain retained
- LRP6 – Sterile alpha motif (SAM)/Pointed domain lost
- PDE3A – Sterile alpha motif (SAM)/Pointed domain retained
- SUFU – Sterile alpha motif (SAM)/Pointed domain retained

Full per-gene detail: [gene\\_reports/etv6/report.md](#)

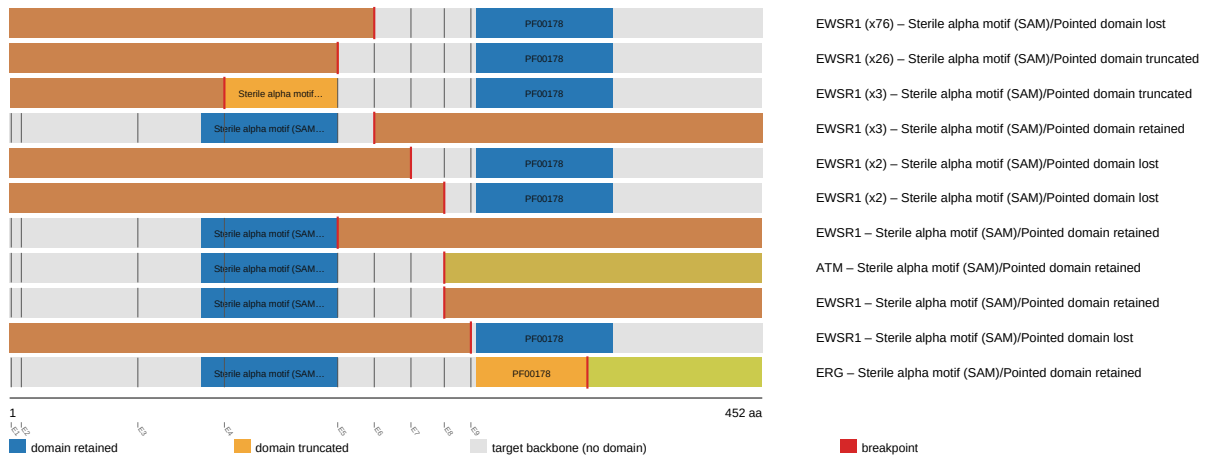
FLI1 (FDR-significant)

FLI1 was analyzed across 118 fusion events, 61.9% in-frame and 5.9% domain-retained. Domain-retention Fisher's exact test  $p=0.988837$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.00820622$  (reaches genome-wide FDR significance).

Reused from the FLI1 individual gene report ([fusion\\_schematic.svg](#)).

### FLI1 fusion-transcript schematic

partner block → breakpoint → retained FLI1 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/fli1/report.md](gene_reports/fli1/report.md)

## RET (Honorable mention, Curated gene config)

RET was analyzed across 194 fusion events, 75.3% in-frame and 92.3% domain-retained. Domain-retention Fisher's exact test  $p=0.000419666$  (raw statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. Genome-wide BH-adjusted  $q=0.096687$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the RET individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

partner block → breakpoint → retained RET portion (domain-colored, exon ticks)

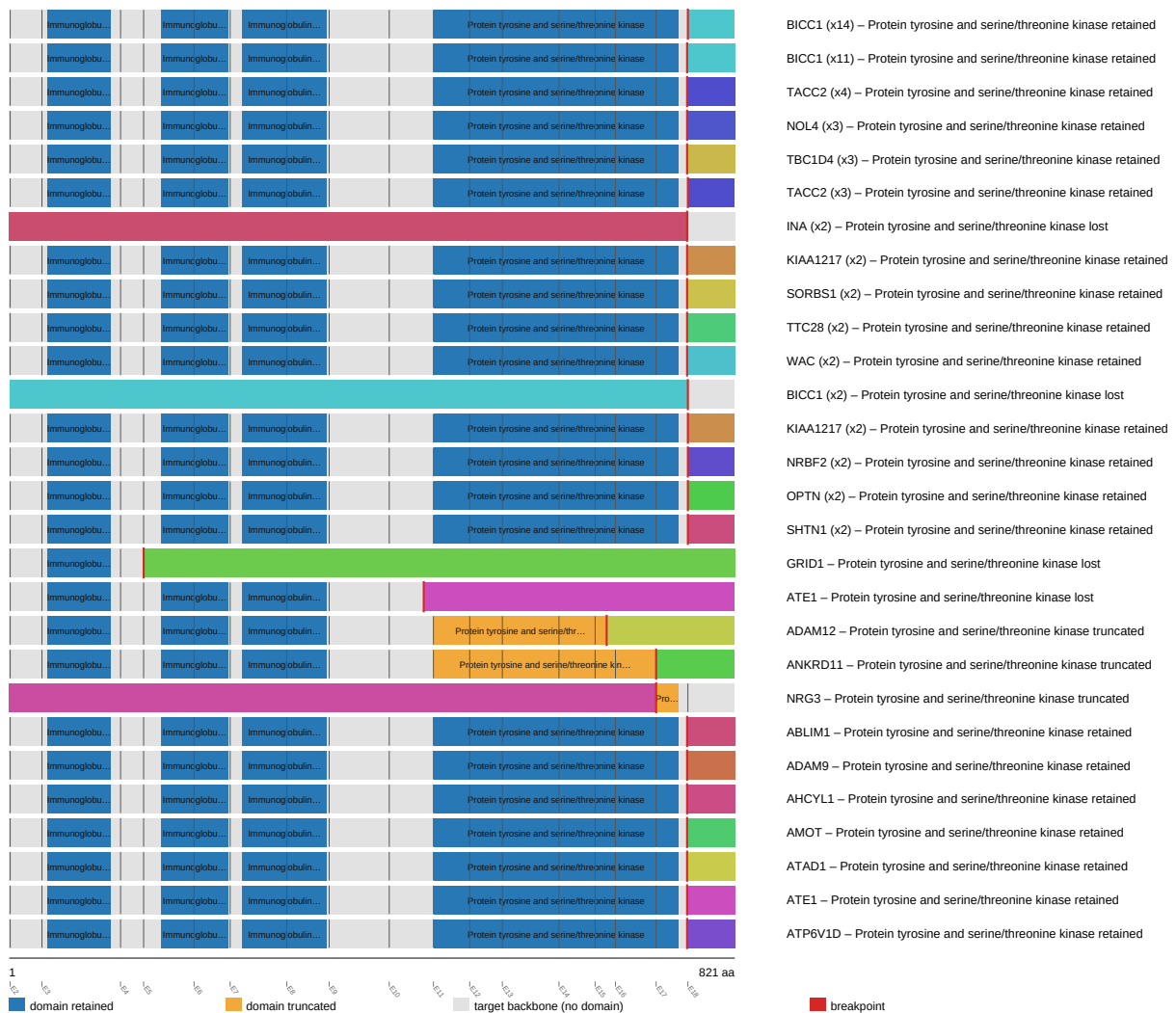


## FGFR2 (Honorable mention)

Reused from the FGFR2 individual gene report (fusion\_schematic.svg).

## FGFR2 fusion-transcript schematic

partner block → breakpoint → retained FGFR2 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/fgfr2/report.md](gene_reports/fgfr2/report.md)

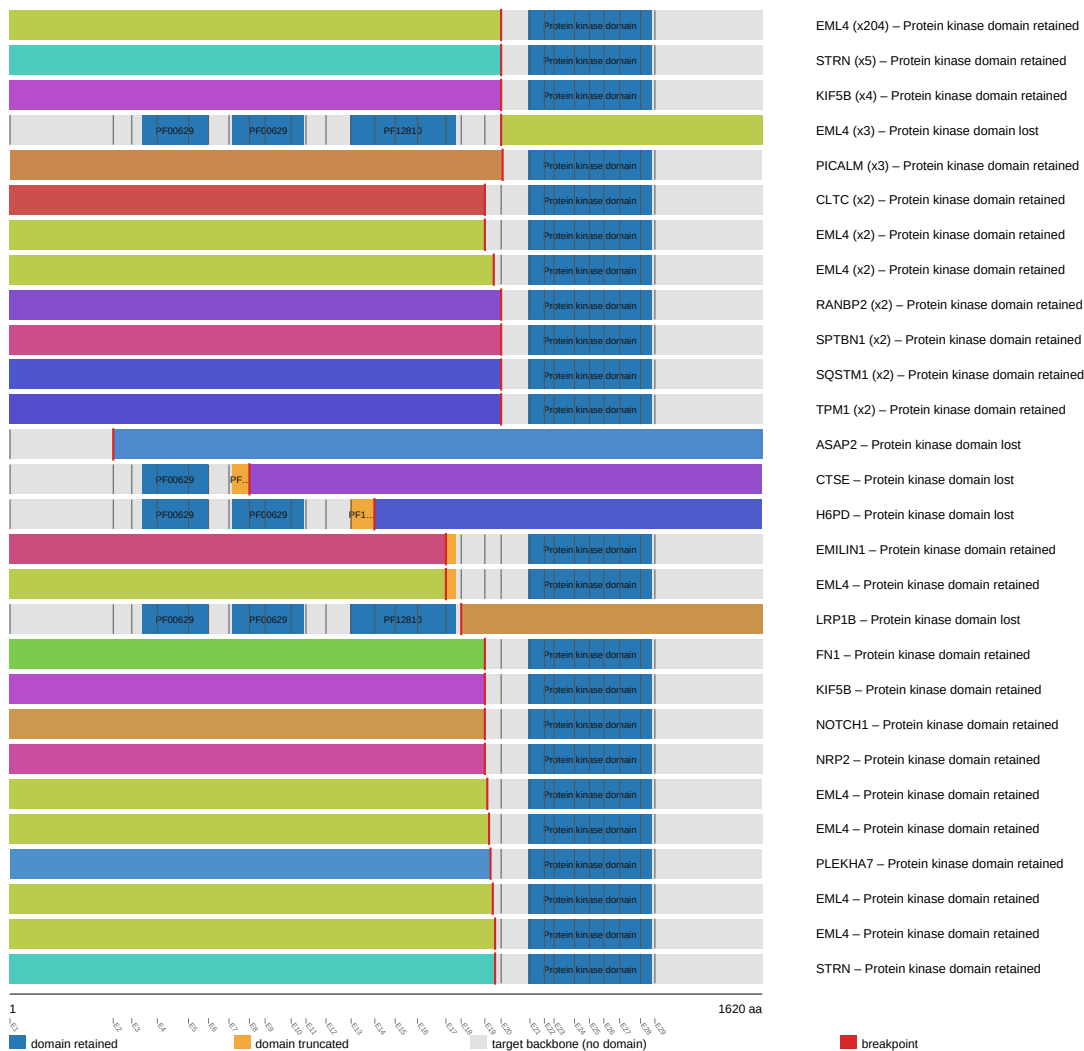
## ALK (Honorable mention, Curated gene config)

ALK was analyzed across 272 fusion events, 83.1% in-frame and 96.0% domain-retained. Domain-retention Fisher's exact test  $p=0.00191661$  (raw statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. Genome-wide BH-adjusted  $q=0.208073$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the ALK individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

## ALK fusion-transcript schematic

partner block → breakpoint → retained ALK portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/alk/report.md](gene_reports/alk/report.md)

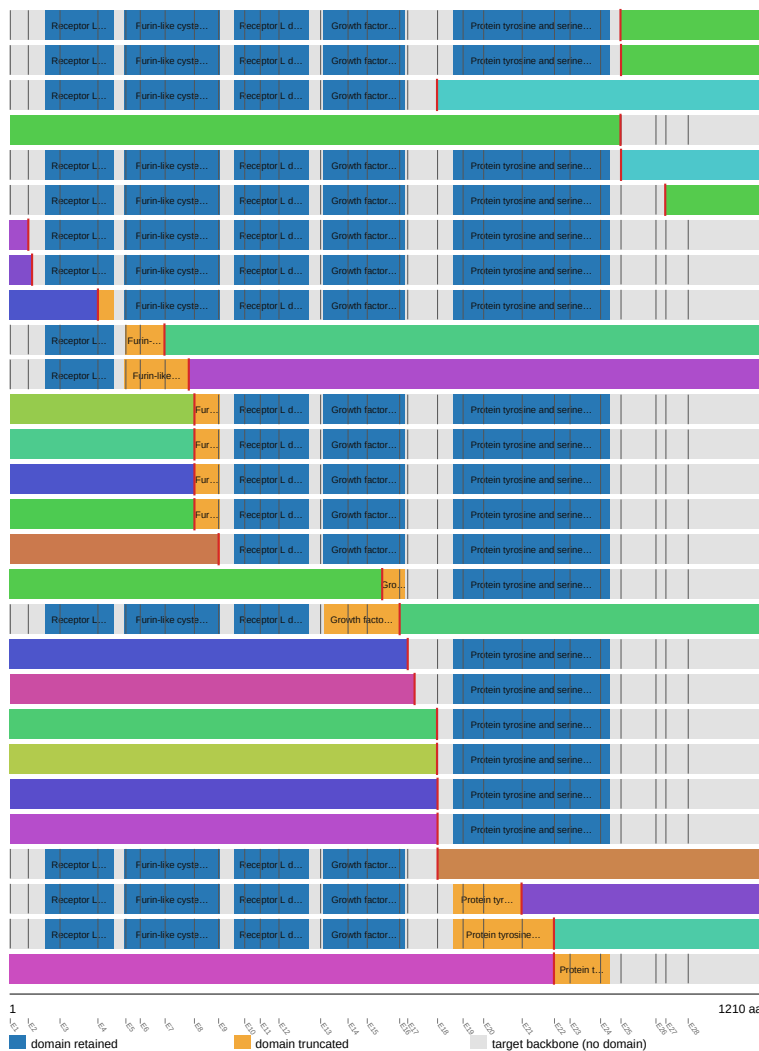
## EGFR (Honorable mention)

EGFR was analyzed across 55 fusion events, 50.9% in-frame and 74.5% domain-retained. Domain-retention Fisher's exact test  $p=0.0114539$  (raw statistically significant at  $\alpha=0.05$ ). The Protein tyrosine and serine/threonine kinase appears to be required for retention. Genome-wide BH-adjusted  $q=0.233522$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the EGFR individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

## EGFR fusion-transcript schematic

partner block → breakpoint → retained EGFR portion (domain-colored, exon ticks)



Showing the top 28 of 41 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

SEPTIN14 (x6) – Protein tyrosine and serine/threonine kinase retained  
 SEPTIN14 (x6) – Protein tyrosine and serine/threonine kinase retained  
 FADD (x2) – Protein tyrosine and serine/threonine kinase lost  
 NIPSNAP2 (x2) – Protein tyrosine and serine/threonine kinase lost  
 RAD51 (x2) – Protein tyrosine and serine/threonine kinase retained  
 SEPTIN14 (x2) – Protein tyrosine and serine/threonine kinase retained  
 PLOD3 – Protein tyrosine and serine/threonine kinase retained  
 ZBPB – Protein tyrosine and serine/threonine kinase retained  
 SCAF4 – Protein tyrosine and serine/threonine kinase retained  
 VOPP1 – Protein tyrosine and serine/threonine kinase lost  
 PCDH15 – Protein tyrosine and serine/threonine kinase lost  
 DDC – Protein tyrosine and serine/threonine kinase retained  
 GARS1 – Protein tyrosine and serine/threonine kinase retained  
 SEL1L – Protein tyrosine and serine/threonine kinase retained  
 TNS3 – Protein tyrosine and serine/threonine kinase retained  
 CSF2RA – Protein tyrosine and serine/threonine kinase retained  
 NIPSNAP2 – Protein tyrosine and serine/threonine kinase retained  
 TUT7 – Protein tyrosine and serine/threonine kinase lost  
 YIF1B – Protein tyrosine and serine/threonine kinase retained  
 PDE1C – Protein tyrosine and serine/threonine kinase retained  
 CADPS – Protein tyrosine and serine/threonine kinase retained  
 NUMA1 – Protein tyrosine and serine/threonine kinase retained  
 EEA1 – Protein tyrosine and serine/threonine kinase retained  
 KIF5B – Protein tyrosine and serine/threonine kinase retained  
 VSTM2A – Protein tyrosine and serine/threonine kinase lost  
 CDH7 – Protein tyrosine and serine/threonine kinase truncated  
 LANCL2 – Protein tyrosine and serine/threonine kinase truncated  
 VWC2 – Protein tyrosine and serine/threonine kinase truncated

Full per-gene detail: [gene\\_reports/egfr/report.md](gene_reports/egfr/report.md)

## BRAF (Honorable mention, Curated gene config)

BRAF was analyzed across 179 fusion events, 84.4% in-frame and 91.1% domain-retained. Domain-retention Fisher's exact test  $p=0.0133676$  (raw statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. Genome-wide BH-adjusted  $q=0.264639$  (does not reach genome-wide FDR significance). The RAS-binding domain and Cysteine-rich domain appear to require loss or disruption rather than retention. Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the BRAF individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

partner block → breakpoint → retained BRAF portion (domain-colored, exon ticks)



### FGFR3 (Honorable mention)

Reused from the FGFR3 individual gene report (fusion\_schematic.svg).

partner block → breakpoint → retained FGFR3 portion (domain-colored, exon ticks)

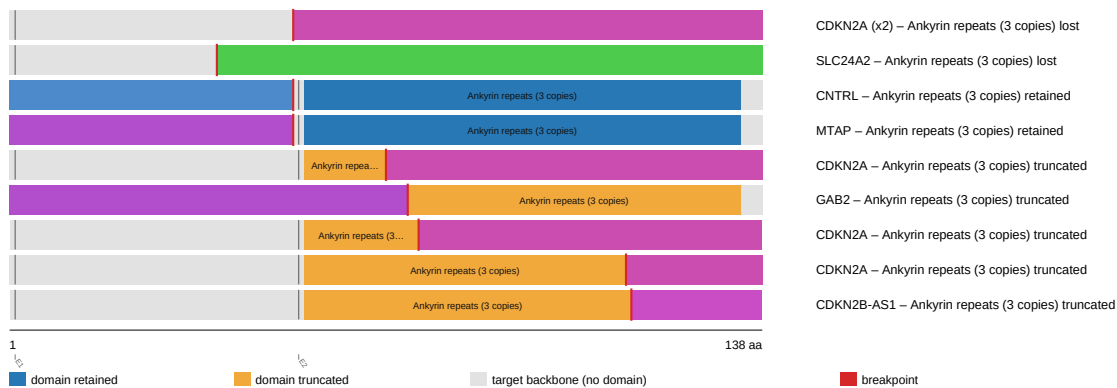


### CDKN2B (Honorable mention)

Reused from the CDKN2B individual gene report (fusion\_schematic.svg).

### CDKN2B fusion-transcript schematic

partner block → breakpoint → retained CDKN2B portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/cdkn2b/report.md](gene_reports/cdkn2b/report.md)

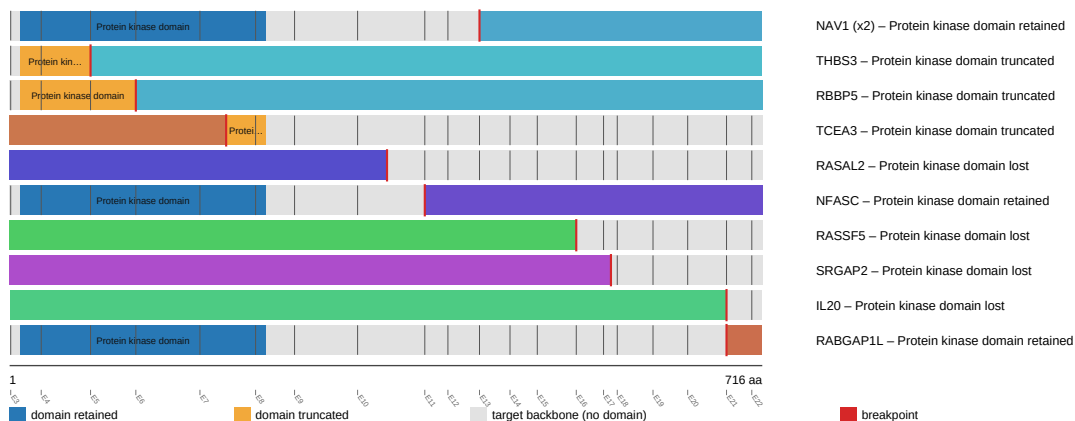
## IKBKE (Honorable mention)

IKBKE was analyzed across 12 fusion events, 25.0% in-frame and 33.3% domain-retained. Domain-retention Fisher's exact test  $p=0.0242424$  (raw statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.321091$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the IKBKE individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

### IKBKE fusion-transcript schematic

partner block → breakpoint → retained IKBKE portion (domain-colored, exon ticks)



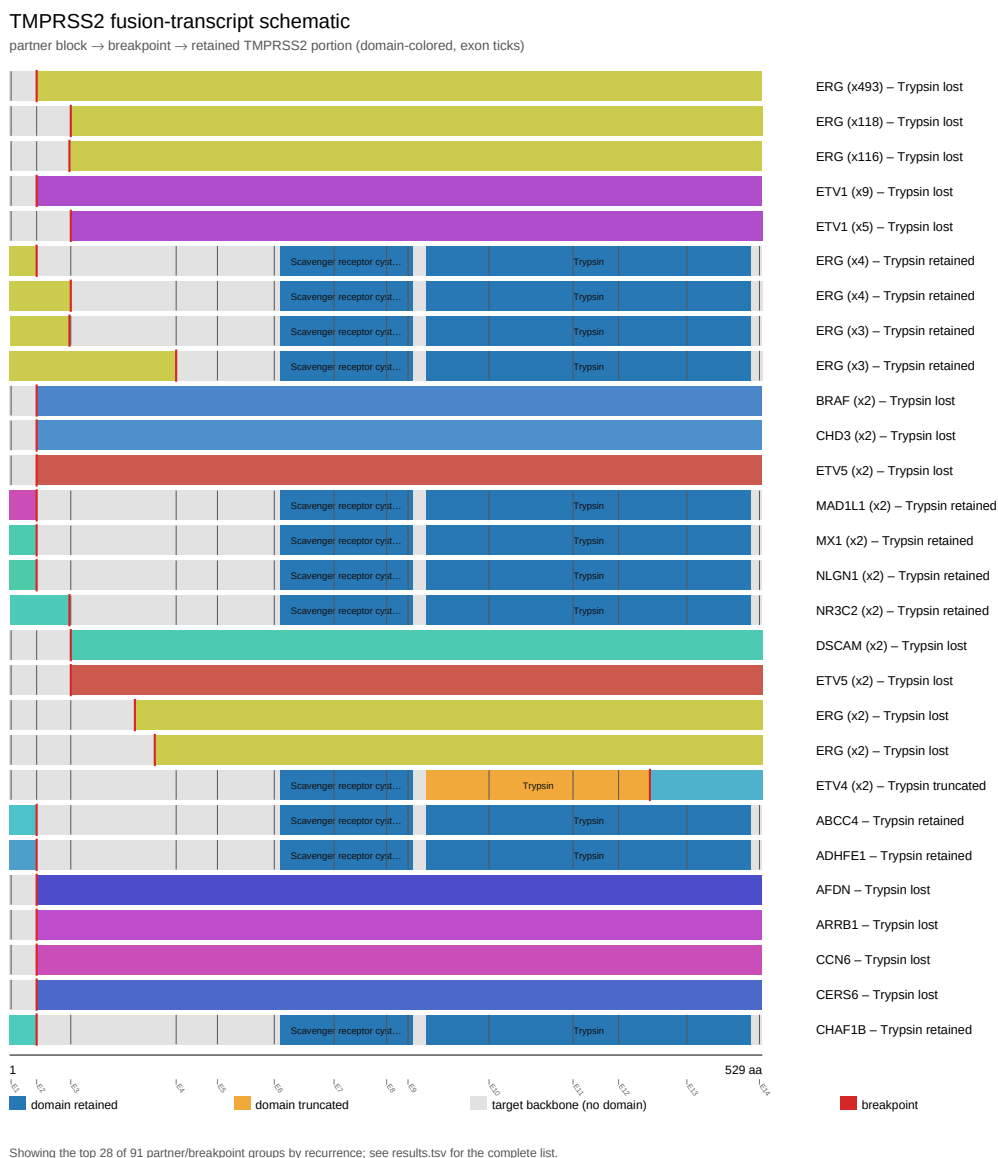
Full per-gene detail: [gene\\_reports/ikbke/report.md](gene_reports/ikbke/report.md)

## TMPRSS2 (Honorable mention)

TMPRSS2 was analyzed across 867 fusion events, 30.0% in-frame and 6.1% domain-retained. Domain-retention Fisher's exact test  $p=0.0292045$  (raw statistically significant at  $\alpha=0.05$ ). The Trypsin appears to be required for retention. Genome-wide BH-adjusted  $q=0.208073$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above

the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the *TMPRSS2* individual gene report (*fusion\_schematic.svg*).



Full per-gene detail: [gene\\_reports/tmpRSS2/report.md](gene_reports/tmpRSS2/report.md)

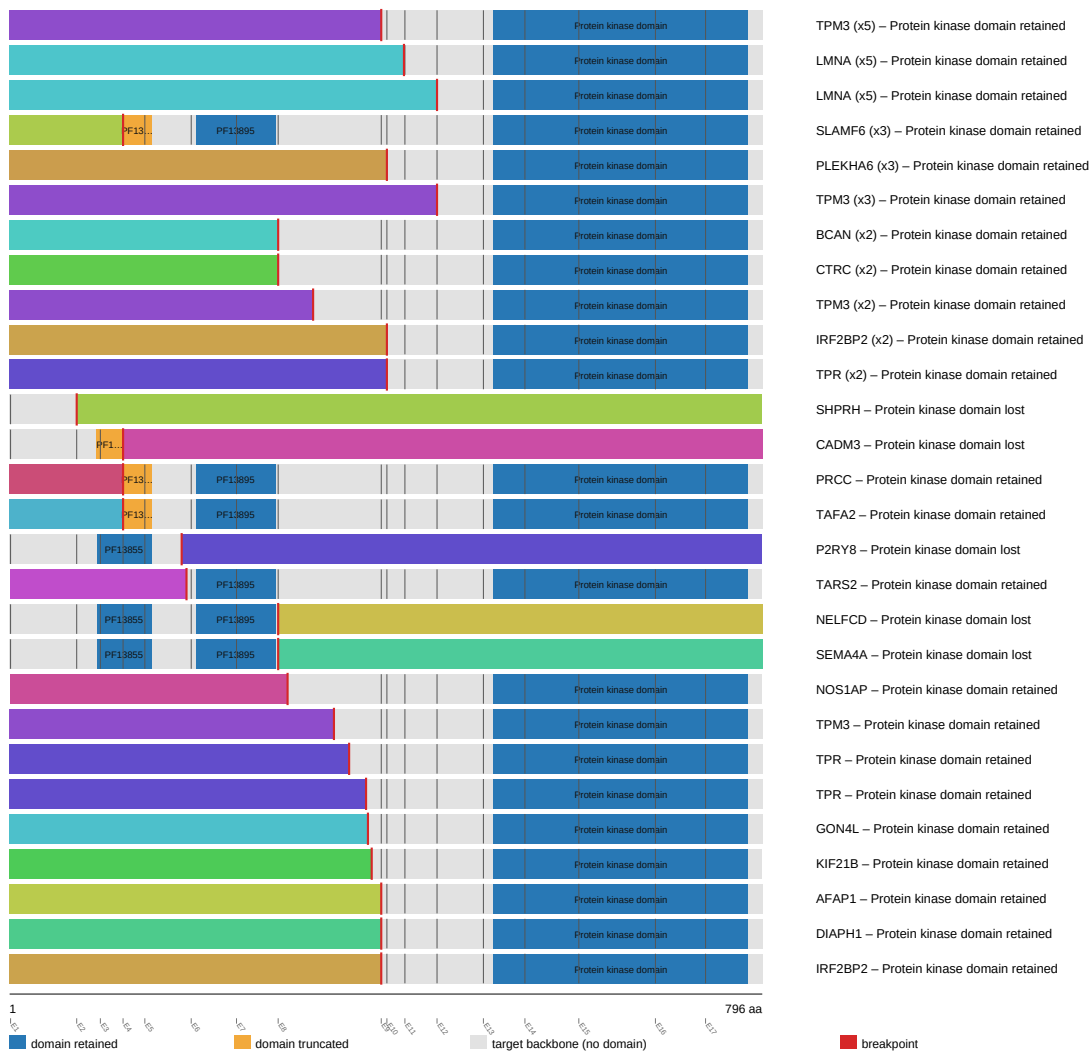
## NTRK1 (Honorable mention, Curated gene config)

NTRK1 was analyzed across 78 fusion events, 41.0% in-frame and 82.1% domain-retained. Domain-retention Fisher's exact test  $p=0.0482628$  (raw statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. Genome-wide BH-adjusted  $q=0.208073$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the *NTRK1* individual gene report (*fusion\_schematic.svg*).

## NTRK1 fusion-transcript schematic

partner block → breakpoint → retained NTRK1 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/ntrk1/report.md](gene_reports/ntrk1/report.md)

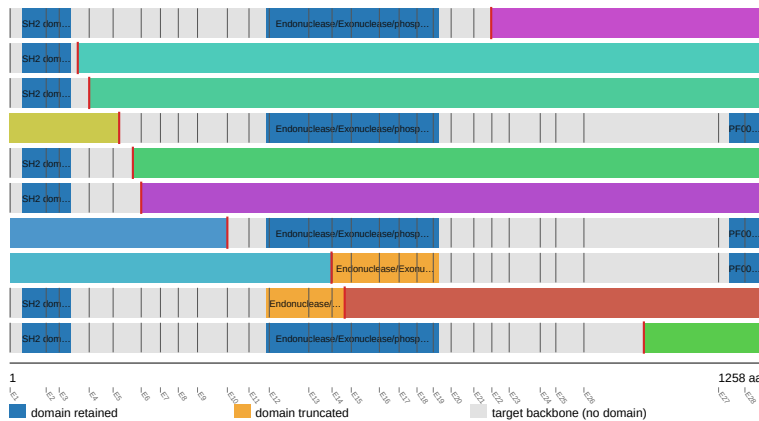
## INPPL1 (Honorable mention)

INPPL1 was analyzed across 14 fusion events, 28.6% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test  $p=0.048951$  (raw statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.468663$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the INPPL1 individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

### INPPL1 fusion-transcript schematic

partner block → breakpoint → retained INPPL1 portion (domain-colored, exon ticks)



FCHSD2 (x4) – Endonuclease/Exonuclease/phosphatase family retained  
 FAT3 – Endonuclease/Exonuclease/phosphatase family lost  
 FOLR2 – Endonuclease/Exonuclease/phosphatase family lost  
 CLPB – Endonuclease/Exonuclease/phosphatase family retained  
 NLRP6 – Endonuclease/Exonuclease/phosphatase family lost  
 GAB2 – Endonuclease/Exonuclease/phosphatase family lost  
 PTCH1 – Endonuclease/Exonuclease/phosphatase family retained  
 SHANK2 – Endonuclease/Exonuclease/phosphatase family truncated  
 ANK2 – Endonuclease/Exonuclease/phosphatase family truncated  
 LRTOMT – Endonuclease/Exonuclease/phosphatase family retained

Full per-gene detail: [gene\\_reports/inppl1/report.md](gene_reports/inppl1/report.md)

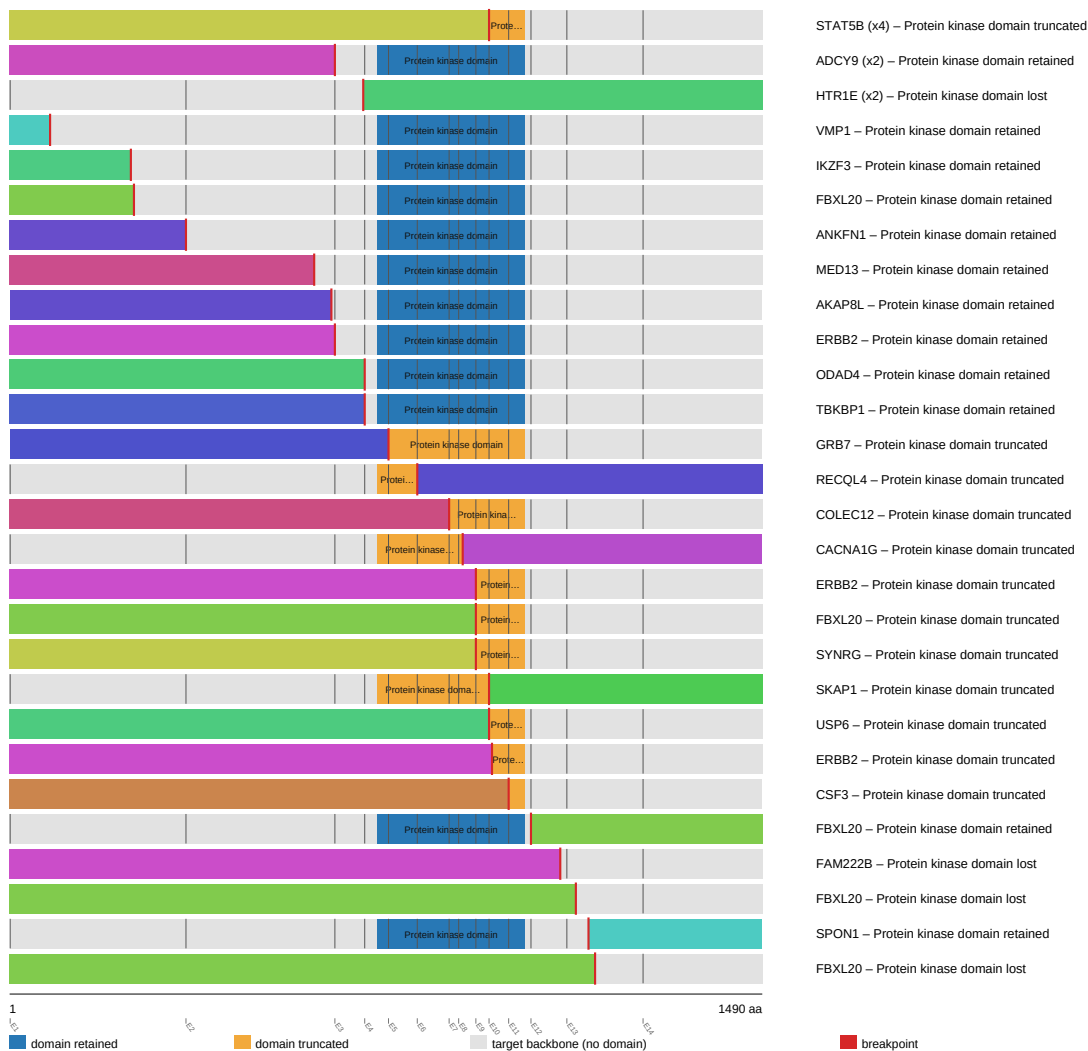
## CDK12 (Honorable mention)

CDK12 was analyzed across 43 fusion events, 16.3% in-frame and 37.2% domain-retained. Domain-retention Fisher's exact test  $p=0.0677006$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.208073$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDK12 individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

## CDK12 fusion-transcript schematic

partner block → breakpoint → retained CDK12 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/cdk12/report.md](gene_reports/cdk12/report.md)

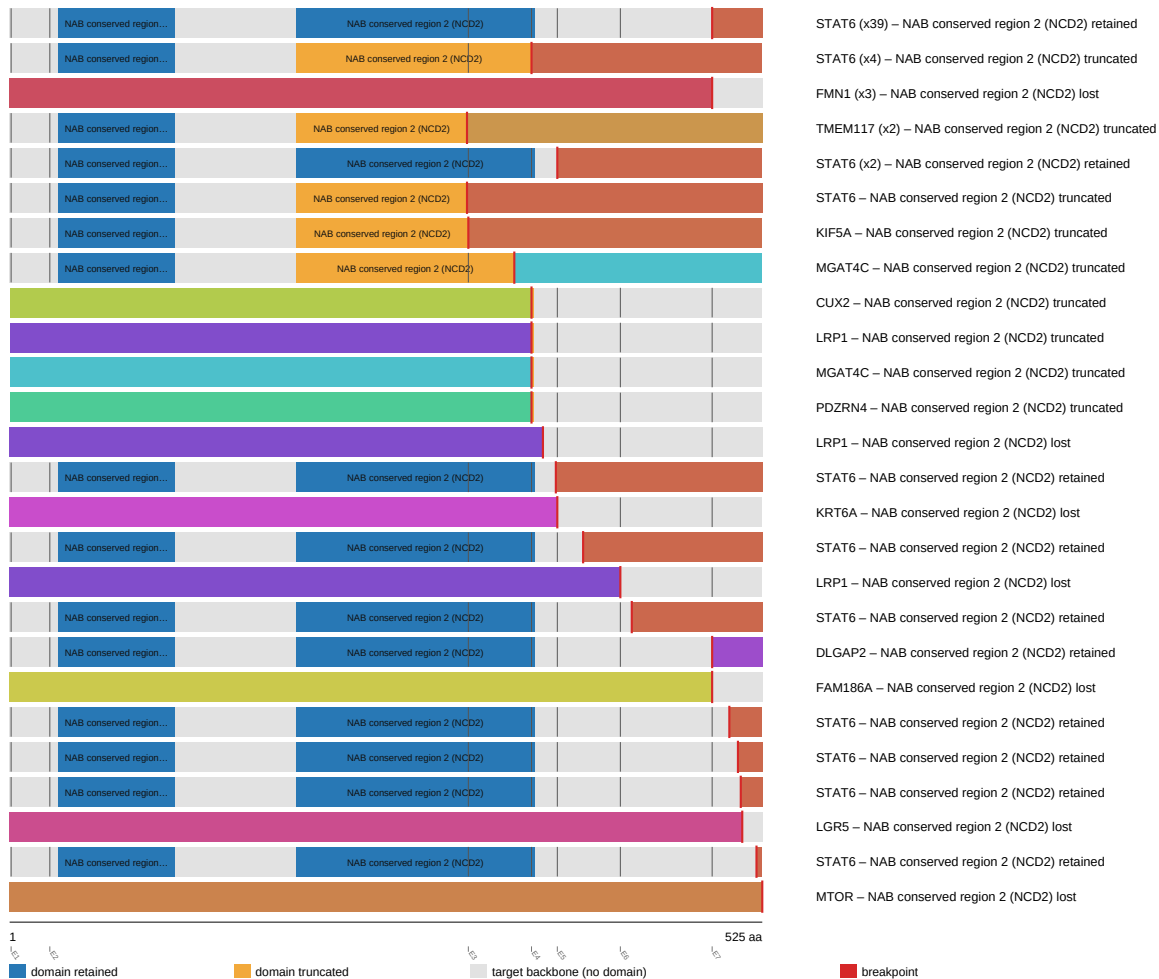
## NAB2 (Honorable mention)

NAB2 was analyzed across 72 fusion events, 34.7% in-frame and 68.1% domain-retained. Domain-retention Fisher's exact test  $p=0.11258$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.208073$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NAB2 individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

## NAB2 fusion-transcript schematic

partner block → breakpoint → retained NAB2 portion (domain-colored, exon ticks)



Full per-gene detail: [gene\\_reports/nab2/report.md](gene_reports/nab2/report.md)

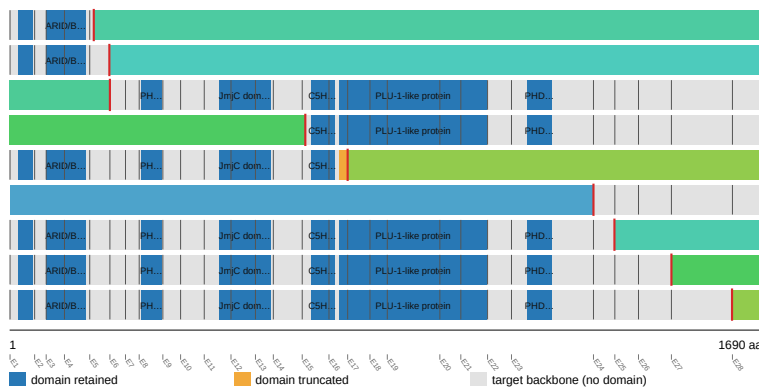
## KDM5A (Honorable mention)

KDM5A was analyzed across 10 fusion events, 30.0% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test  $p=0.119048$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.898036$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the KDM5A individual gene report ([fusion\\_schematic.svg](fusion_schematic.svg)).

### KDM5A fusion-transcript schematic

partner block → breakpoint → retained KDM5A portion (domain-colored, exon ticks)



OSBPL8 – PLU-1-like protein lost  
STAT5A – PLU-1-like protein lost  
CACNA1C – PLU-1-like protein retained  
BMAL2 – PLU-1-like protein retained  
NINJ2 – PLU-1-like protein truncated  
COLGALT1 – PLU-1-like protein lost  
DIPK1A – PLU-1-like protein retained  
DHX37 – PLU-1-like protein retained  
NINJ2 – PLU-1-like protein retained

Full per-gene detail: [gene\\_reports/kdm5a/report.md](http://gene_reports/kdm5a/report.md)

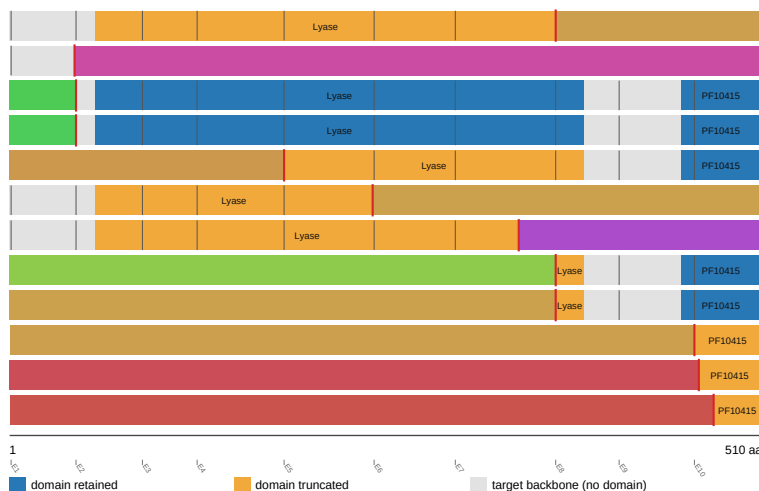
## FH (Honorable mention)

FH was analyzed across 16 fusion events, 6.2% in-frame and 12.5% domain-retained. Domain-retention Fisher's exact test  $p=0.133333$  (raw not statistically significant at  $\alpha=0.05$ ). Genome-wide BH-adjusted  $q=0.686319$  (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted  $q$ -value at or above the significance threshold), but ranks highly by raw  $p$ -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FH individual gene report ([fusion\\_schematic.svg](http://fusion_schematic.svg)).

### FH fusion-transcript schematic

partner block → breakpoint → retained FH portion (domain-colored, exon ticks)



DGKH (x4) – Lyase truncated  
PDE1C – Lyase lost  
RYS2 – Lyase retained  
unknown – Lyase retained  
NOTCH1 – Lyase truncated  
RGS7 – Lyase truncated  
FMN2 – Lyase truncated  
ARID4B – Lyase truncated  
RGS7 – Lyase truncated  
DGKH – Lyase lost  
MAFTRR – Lyase lost  
SLC35F3 – Lyase lost

Full per-gene detail: [gene\\_reports/fh/report.md](http://gene_reports/fh/report.md)

## Discussion

- Cross-gene Benjamini-Hochberg FDR correction was applied jointly across the 359 genes that produced at least one computable  $p$ -value in this scan. This reduces false-positive findings relative to testing each gene in isolation, but is a conservative correction: a real per-gene effect can fail to reach the  $q < 0.05$  threshold once

corrected across that full p-value-bearing gene set.

- 4 gene(s) used hand-curated gene configurations, while 520 gene(s) were auto-configured from Genome Nexus canonical-transcript/Pfam-domain data using a kinase/catalytic-keyword heuristic to select the tracked domain; auto-configured domains have not been manually verified the way hand-curated ones have.
- Data were retrieved live from the public msk\_impact\_50k\_2026 cBioPortal study as of 2026-09-04T19:24:30.753442+00:00. Some cBioPortal cohorts (e.g. actively accruing clinical-sequencing panels such as MSK-IMPACT) are updated periodically, so exact counts could shift if this scan is re-run later against such a cohort.
- All findings in this manuscript are computational, hypothesis-generating candidate evidence from bioinformatic analysis of public cohort data, not validated clinical calls; a gene's presence here is not a therapeutic or diagnostic recommendation.

## Appendix: per-gene report index

| Gene    | Tier                                   | Config source | N events | Best q    | Full report                    |
|---------|----------------------------------------|---------------|----------|-----------|--------------------------------|
| NTRK3   | FDR-significant                        | auto          | 58       | 8.165e-09 | gene_reports/ntrk3/report.md   |
| ROS1    | FDR-significant                        | auto          | 122      | 0.0002213 | gene_reports/ros1/report.md    |
| ETV6    | FDR-significant                        | auto          | 90       | 0.002333  | gene_reports/etv6/report.md    |
| FLI1    | FDR-significant                        | auto          | 118      | 0.008206  | gene_reports/fli1/report.md    |
| RET     | Honorable mention; Curated gene config | curated       | 194      | 0.09669   | gene_reports/ret/report.md     |
| FGFR2   | Honorable mention                      | auto          | 136      | 0.09669   | gene_reports/fgfr2/report.md   |
| ALK     | Honorable mention; Curated gene config | curated       | 272      | 0.2081    | gene_reports/alk/report.md     |
| EGFR    | Honorable mention                      | auto          | 55       | 0.2335    | gene_reports/egfr/report.md    |
| BRAF    | Honorable mention; Curated gene config | curated       | 179      | 0.2646    | gene_reports/braf/report.md    |
| FGFR3   | Honorable mention                      | auto          | 152      | 0.2081    | gene_reports/fgfr3/report.md   |
| CDKN2B  | Honorable mention                      | auto          | 12       | 0.2081    | gene_reports/cdkn2b/report.md  |
| IKBKE   | Honorable mention                      | auto          | 12       | 0.3211    | gene_reports/ikbke/report.md   |
| TMPRSS2 | Honorable mention                      | auto          | 867      | 0.2081    | gene_reports/tmpRSS2/report.md |
| NTRK1   | Honorable mention; Curated gene config | curated       | 78       | 0.2081    | gene_reports/ntrk1/report.md   |
| INPPL1  | Honorable mention                      | auto          | 14       | 0.4687    | gene_reports/inppl1/report.md  |
| CDK12   | Honorable mention                      | auto          | 43       | 0.2081    | gene_reports/cdk12/report.md   |
| NAB2    | Honorable mention                      | auto          | 72       | 0.2081    | gene_reports/nab2/report.md    |
| KDM5A   | Honorable mention                      | auto          | 10       | 0.898     | gene_reports/kdm5a/report.md   |
| FH      | Honorable mention                      | auto          | 16       | 0.6863    | gene_reports/fh/report.md      |